

KARIM AYMAN ABD-ELMAQSOOD KHALIL

BIM Modeler & Technical Office Draftsman — Civil & Architectural

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PROFESSIONAL SUMMARY

BIM Modeler & Technical Office Draftsman with 5+ years of experience across both civil and architectural drafting disciplines on Egypt's largest infrastructure projects, including the El Dabaa Nuclear Power Plant (Units 1, 3 & 4) — one of the most technically demanding projects in Africa. Built a solid foundation in traditional technical office procedures, then fast-tracked BIM mastery by introducing Revit workflows to projects lacking digital infrastructure, closing design gaps and optimizing interdisciplinary coordination. Specialized in producing precision shop drawings, rebar detailing, and waterproofing details under strict GOST/Rosatom and NPPA nuclear compliance standards, with a track record of high first-submission consultant approval rates. Proficient in Revit, AutoCAD, and Civil 3D, delivering accurate, coordinated, and audit-ready As-Built drawings across civil and architectural scopes.

SKILLS

BIM & CAD Tools: AutoCAD (Advanced) | BIM Modeling | Revit (Architecture & Structure) | Autodesk Construction Cloud (ACC) | Civil 3D | ArcGIS.

Technical Office & Documentation: Shop Drawings | Rebar Detailing | Bar Bending Schedule (BBS) | As-Built Drawings | RFI Resolution | P&ID Drafting | Quantity Takeoff (QTO).

Standards & Compliance: ISO 19650 | BIM Execution Plan (BEP) Compliance | LOD 200-500 | GOST | Rosatom | NPPA Nuclear Standards

Coordination: MEP Coordination | Clash Detection | Worksets Management | Naming Conventions .

Office Tools: Microsoft Office (Excel, Word, PowerPoint) | Google Workspace.

PROFESSIONAL EXPERIENCE

BIM Modeler & Technical Office Draftsman

Feb 2023 – Present

Hassan Allam Construction (HAC) | El Dabaa, Egypt

Project: El Dabaa Nuclear Power Plant — Units 3 & 4

Client: Nuclear Power Plants Authority (NPPA)

Main Contractor: ASE / Rosatom

- Produced **500+ shop drawings** from Detailed Design Drawings (DDD) for waterproofing and structural systems in nuclear reactor foundations, achieving high first-submission approval rates under strict GOST/Rosatom and NPPA compliance standards.
- Initiated and developed **3D models** for structural elements and buildings to facilitate clash detection (LOD 350) and early identification of site issues, reducing error rates and streamlining preparation of accurate As-Built Drawings (LOD 500) and use BIM Execution Plan.
- Prepared detailed reinforcement Shop Drawings and Bar Bending Schedules (BBS) for critical structural elements in nuclear reactor foundations, directly from DDD packages, ensuring full adherence to international nuclear safety codes, GOST, Rosatom, and NPPA standards.
- Translated Work Execution Plans (WEP) into accurate bay boundaries, construction joints, and sequential waterproofing interfaces within shop drawing documentation.
- Delivered comprehensive as-built drawings for steel reinforcement, waterproofing systems, and repair stages, supporting NPPA quality audits and formal handover records.
- Resolved site RFIs through coordinated drawing revisions with Egyptian and Russian engineering teams, preserving design intent across all documentation stages.
Applied BIM methodology (LOD 200-300) for conceptual modeling and automated quantity takeoff (QTO), improving accuracy of tendering and cost estimation workflows.

Technical Office Draftsman

Mar 2022 – Feb 2023

MEGA Construction | El Dabaa, Egypt

Project: El Dabaa Nuclear Power Plant — Unit 1

Client: Nuclear Power Plants Authority (NPPA)

Main Contractor: ASE / Rosatom

- Extracted and developed **150+ high-precision structural Shop Drawings** and complex Rebar configurations directly from DDD packages, resolving dimensional mismatches and accelerating consultant approvals by 20%.
- Drafted specialized execution drawings for 40+ massive concrete bays in strict accordance with the Work Execution Plan (WEP), integrating sensor locations, feed points, and construction joints to prevent site execution delays.
- Optimized and revised Rebar Shop Drawings under tight deadlines, converting approved technical solutions for 80+ complex reinforcement clashes into execution-ready layouts, eliminating material waste.
- Produced detailed structural sections, elevations, and construction layouts using AutoCAD, managing 200+ Red-Line Markups and emergency design changes from site RFIs while maintaining strict CAD Standards and Version Control.
- Coordinated closely with the Russian engineering team and surveying engineers to deliver final As-Built Drawings for Unit 1, utilizing verified field measurements and achieving a 5% rejection rate during final handover and inspection.

Technical Office Draftsman

May 2021 – Mar 2022

Orascom Construction PLC | Giza, Egypt

Project: Abu Rewash Wastewater Treatment Plant

Client: Construction Authority for Potable Water and Wastewater (CAPW)

Main Contractor: Orascom Construction PLC & Aqualia

- Drafted P&ID diagrams for water treatment processes in coordination with mechanical engineering inputs.
- Produced detailed reinforced concrete structural drawings and shop drawings for water tanks and plant facilities.
- Developed mechanical equipment layouts and piping route drawings with supporting interdisciplinary coordination.
- Documented final AsBuilt conditions for all piping and structural elements, enabling a reduction in RFI submissions by 25% during the closeout phase.

CERTIFICATIONS & TRAINING

- Revit Architecture Essentials — Autodesk Authorized Training Center (June 2026).
- Revit Structure Essentials — Autodesk Authorized Training Center (April 2026).
- Revit Quantity Takeoff — Autodesk Authorized Training Center (January 2026).
- Introduction to BIM — Autodesk Authorized Training Center (December 2025).
- AutoCAD — Advanced Drafting & Shop Drawing (2020).

EDUCATION

Bachelor of Arts — Geography & GIS & Cartography

Major: Geographic Information Systems (GIS), Cartography & Surveying.

Tanta University | Graduated 2021.

LANGUAGES

Arabic — Native

English — Fluent, professional working proficiency across multinational project environments (Egyptian, Russian & international teams at El Dabaa NPP).